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# P4 — Explaining Motion

OCR GCSE Physics B (J259) — 21st Century Science · Paper 2 (Depth in Physics)

## P4.1 Scalars and Vectors

A **scalar** quantity has magnitude (size) only — e.g. distance, speed, mass, energy, time.
A **vector** quantity has magnitude *and* direction — e.g. displacement, velocity, acceleration, force, momentum, weight.

| Scalar | Vector |
|---|---|
| Distance — total path length | Displacement — straight-line distance *and* direction from start |
| Speed — how fast | Velocity — speed in a stated direction |

## P4.2 Speed, Velocity and Acceleration

**Speed**: $v = \text{distance} / \text{time} \quad (\text{m/s})$

**Acceleration**: $a = \Delta v / t = (v - u) / t \quad (\text{m/s}^2)$

$u$ = initial velocity, $v$ = final velocity

**SUVAT**: $v^2 = u^2 + 2\,a\,s$

$(\text{m/s})^2 = (\text{m/s})^2 + 2\,(\text{m/s}^2)(\text{m})$

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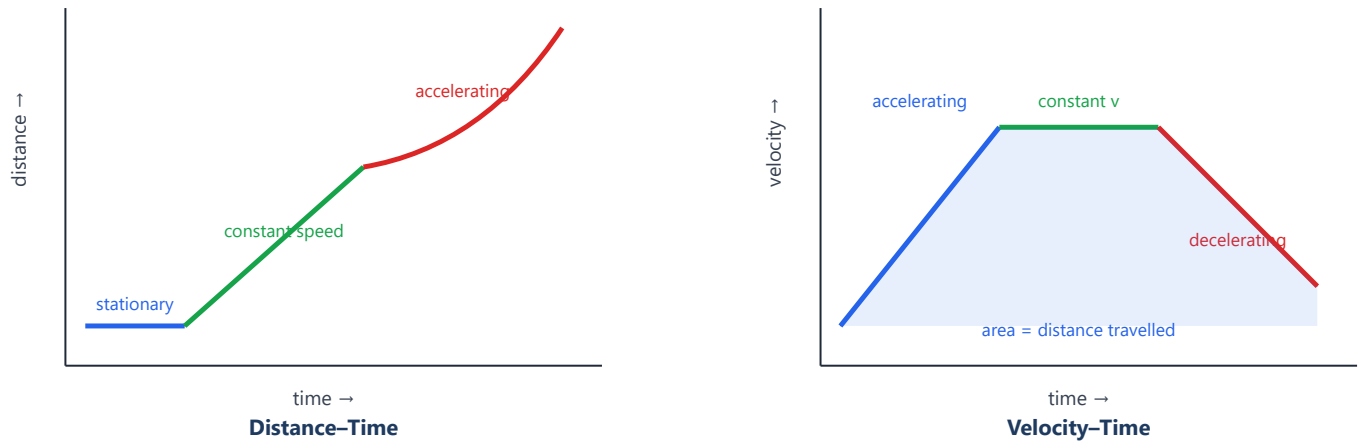
## Typical Speeds

| Situation | Typical speed |
|---|---|
| Walking | 1.5 m/s |
| Running | 3 m/s |
| Cycling | 6 m/s |
| Car (urban) | 13 m/s (30 mph) |
| Car (motorway) | 31 m/s (70 mph) |
| Train | 55 m/s |
| Cruising aeroplane | 250 m/s |
| Sound in air (20 °C) | 340 m/s |

**Worked Example:** A car accelerates from 10 m/s to 25 m/s in 5 s. Find its acceleration.
a = (v − u)/t = (25 − 10)/5 = **3 m/s²**

# P4.3 Distance–Time and Velocity–Time Graphs

**Distance–Time and Velocity–Time Graphs**



- On a **distance–time graph**, the *gradient* gives the **speed**.
- On a **velocity–time graph**, the *gradient* gives the **acceleration** and the *area under the line* gives the **distance travelled**.

## P4.4 Forces

A **force** is a push or a pull. Forces are vectors, measured in **newtons (N)**. Forces can change an object's shape, speed or direction.

### Contact and Non-Contact Forces

- **Contact**: friction, tension, normal reaction, air resistance, drag, upthrust.
- **Non-contact**: gravity, magnetic force, electrostatic force.

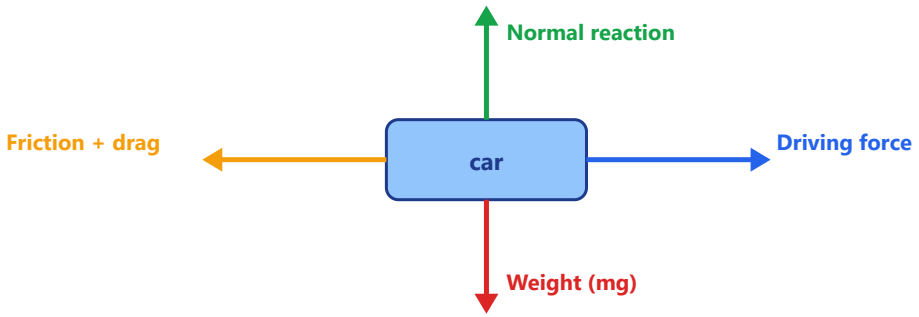
### Weight and Mass

$$W = m \times g \quad (N) = (kg) \times (N/kg)$$

On Earth, g ≈ 9.8 N/kg. *Mass* (kg) is the amount of matter; *weight* (N) is the gravitational force acting on that mass.

### Free-Body Diagrams

**Free-Body Diagram — Car at Constant Speed**




At constant velocity, forces balance: driving force = friction + drag.

## P4.5 Newton's Laws of Motion

**Newton's 1st Law:** an object at rest stays at rest, and an object moving in a straight line continues at constant velocity, *unless* acted on by a resultant (net) force.

**Newton's 2nd Law:** the resultant force on an object equals its mass × its acceleration.

$$F = m \times a \quad (N) = (kg) \times (m/s^2)$$

**Newton's 3rd Law:** when object A exerts a force on object B, object B exerts an equal and opposite force on A. These form a *pair* — same size, opposite direction, acting on two different bodies.

**Worked Example:** A 1200 kg car has a resultant force of 3000 N. What is its acceleration?
a = F / m = 3000 / 1200 = **2.5 m/s²**

## P4.6 Inertia and Inertial Mass

**Inertia** is the tendency of an object to keep doing what it is already doing (staying at rest, or continuing in a straight line at constant velocity). A more massive object has more inertia — it is harder to accelerate.

The ratio **F / a** in Newton's 2nd Law is called the **inertial mass**.

## P4.7 Momentum

**Momentum (p)** = mass × velocity. Unit: **kg m/s**. Momentum is a *vector*.

$$p = m \times v$$

Momentum is always **conserved** in a closed system (where no external forces act): *total momentum before = total momentum after*.

**Worked Example:** A 0.5 kg ball moving at 4 m/s hits a stationary 1.5 kg ball and they stick together. Find their common velocity.
Before: p = 0.5 × 4 + 1.5 × 0 = 2 kg m/s
After: p = (0.5 + 1.5) v = 2v. So v = 1 m/s.

### Force as Rate of Change of Momentum

$$F = \Delta p / \Delta t = m(v - u) / t$$

A smaller force is needed if the change of momentum happens over a **longer time**. This is why airbags, crumple zones, cycle helmets, crash mats, and seat belts all *increase stopping time* — reducing the peak force on the body during a collision.

## P4.8 Stopping Distance

**Stopping distance** = **thinking distance** + **braking distance**.
Thinking distance: distance travelled during the driver's reaction time.
Braking distance: distance travelled while the brakes decelerate the car to rest.

| Affects thinking distance | Affects braking distance |
|---|---|
| Speed | Speed |
| Tiredness | Road conditions (wet/icy/gravel) |
| Alcohol/drugs | Tyre condition (tread, pressure) |

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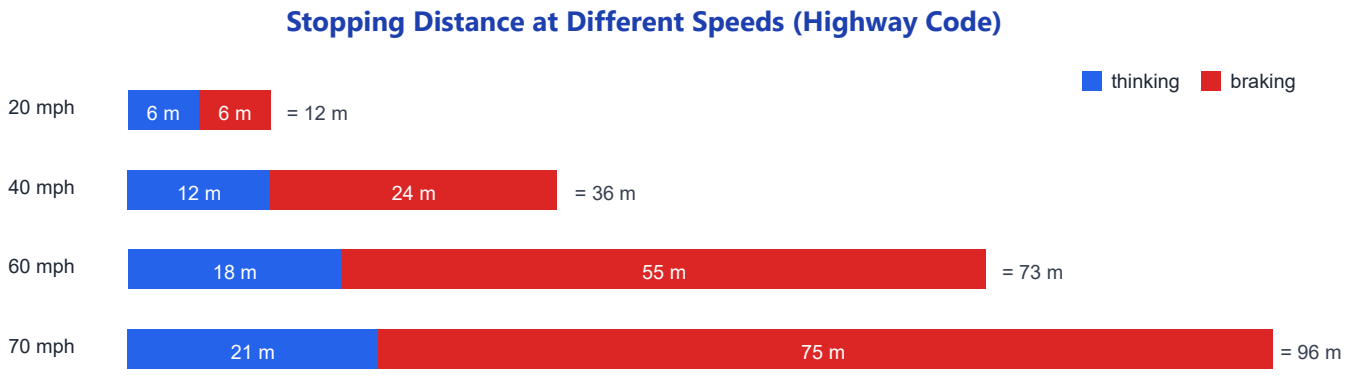
| | |
|---|---|
| Distractions (phone) | Brake condition |




*Braking distance grows with $v^2$ — doubling speed roughly quadruples braking distance.*

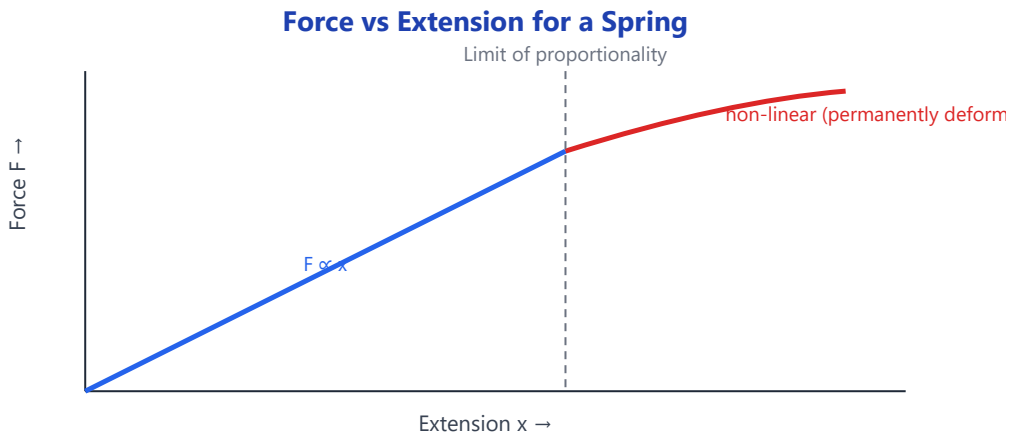
### Energy Dissipated in Braking

During braking, the car's kinetic energy is transferred to **thermal energy** in the brake discs, pads and tyres (via friction). Because $E_k = ½ m v^2$, doubling the speed means *four times* as much energy to dissipate, so the braking distance also roughly quadruples.

## P4.9 Hooke's Law (Forces and Elasticity)

The **extension** of an elastic object (spring, rubber band, wire) is directly proportional to the force applied, provided the limit of proportionality is not exceeded.

$$F = k \times x \quad (N) = (N/m) \times (m)$$




**Worked Example:** A spring of constant 40 N/m is stretched by 0.15 m. Find the elastic PE stored.

$E_e = ½ k x^2 = ½ \times 40 \times 0.15^2 = ½ \times 40 \times 0.0225 =$ **0.45 J**

## P4.10 Moments (Turning Forces)

A **moment** is the turning effect of a force about a pivot.

**Moment (N m) = Force (N) × perpendicular distance from pivot (m)**

**Exam Tip:** **Principle of moments** (for a body in equilibrium): total clockwise moments = total anticlockwise moments.

**Worked Example:** A see-saw is 4 m long, with the pivot in the middle. A 50 kg child sits 1.5 m left of the pivot. Where must a 30 kg child sit to balance it?

Left moment = 50 × 9.8 × 1.5 = 735 N m

Right moment = 30 × 9.8 × d = 735 → d = 735 / (30 × 9.8) = **2.5 m** to the right.